

Cloudy Night

Activating Park Space by Illuminated Interactivity

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Executive Summary

Cloudy Night is an interactive lighting and data collecting prototype. The device was developed to provide illuminated feedback loops in low light conditions throughout the city while collecting unidentifiable data on its participants. The intent of the project was to assess if interactive lighting could enhance public park engagement and safety. The first deployment occurred on May 4th, 2019 at Starlight Park in the Bronx, New York.

The stakeholders of the project were the Bronx River Alliance with special consideration given to the people and parks they serve. After a 3 hour deployment of the prototype in the afternoon and evening, the collected data was analyzed for quality, focusing on interpretability and inference. It was found that both directionality of human movement and the presence of children could be roughly approximated. The presence of children was used as a proxy for the perception of public safety, where it was assumed that mothers will not bring their children into environments that are perceived to be unsafe.

Although this was assumed, and a rigorous analysis was not conducted to assess the number of occurrences of each event, it was more important that this project demonstrated that capturing these areas of interest was possible. It was determined that the Cloudy Night prototype was a useful device for creating interactive illuminances, while assessing rough approximations of public park safety and human movement through both directionality and the presence of children.



Introduction

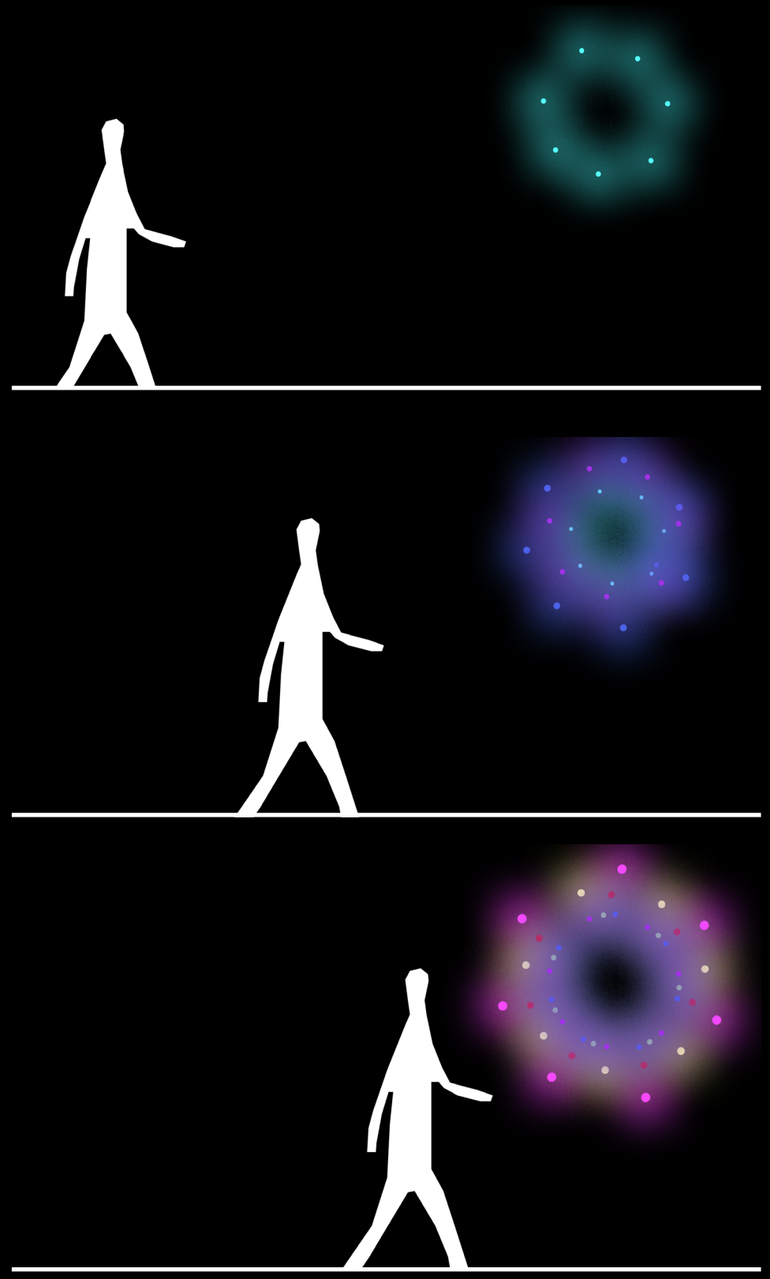
People go to the park to relax, to walk, to talk, to engage with one another. Cloudy Night sought to better understand these nuances of park usage and public safety, while enhancing the user experience with an empirical way to measure it.

The project focused on illuminate low light areas with fun, interactive, and playful feedback loops. It assessed if these mediums of interaction might work to enhance park engagement and the perception of public safety.

The presence of children was used as a proxy for the perception of public safety, where it was assumed that mothers will not bring their children into environments that are perceived to be unsafe.

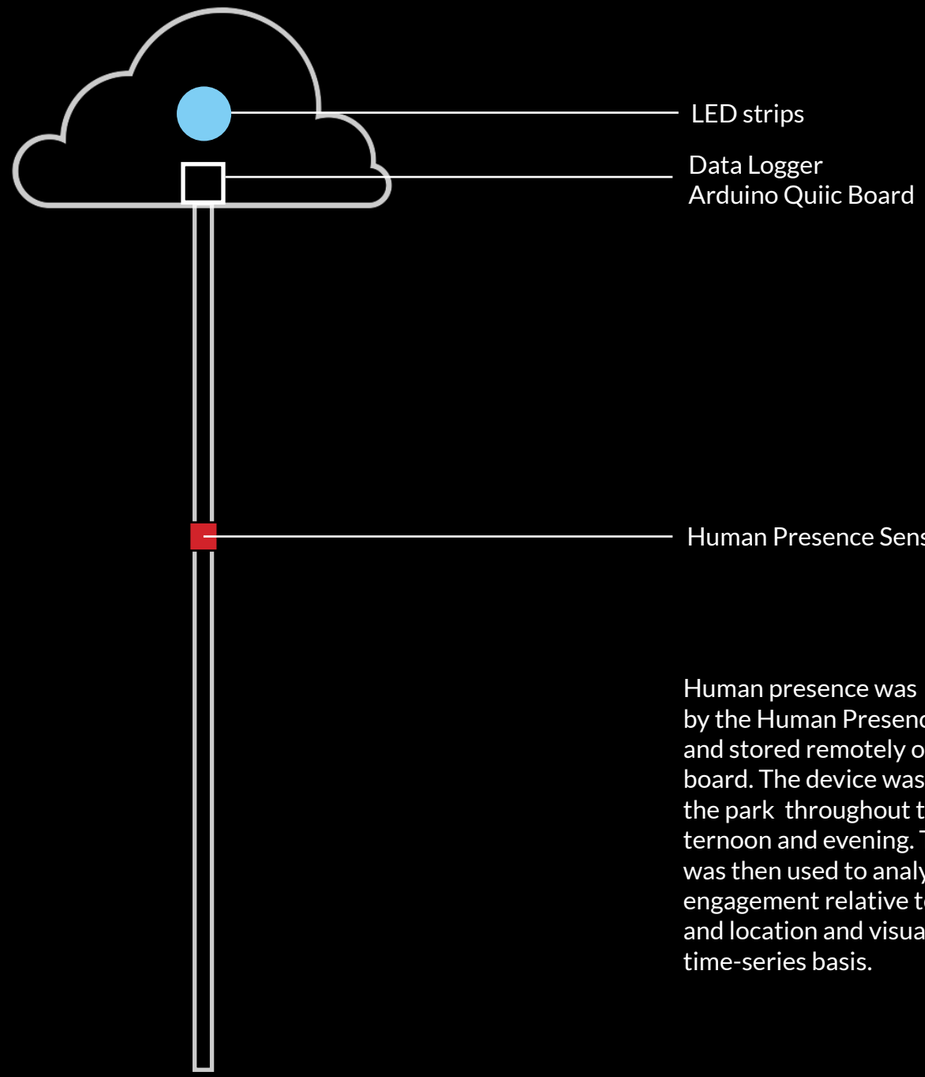
This project asked the following: Can illuminating low lit park spaces with interactive lighting devices help to make people feel safer? Can illuminated interactivity enhance park engagement? Can the detection of children interacting with the device be used as a proxy for public safety?

Interaction



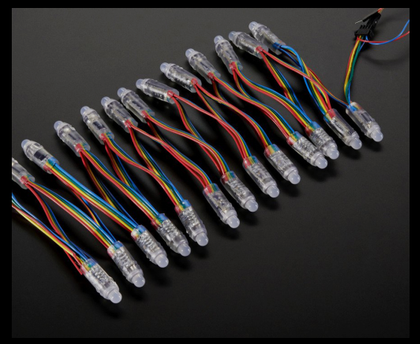
The interaction scale of the project is approximately 5-meters around the installation. Sensors were embedded in the device and applied in locations with low afternoon and evening illuminances with potentially low visual access throughout the park. When people come near the cloud, the light inside it was programmed to change color, providing feedback to reflect human presence.

Technology



Human presence was recorded by the Human Presence Sensor and stored remotely on the SD board. The device was placed in the park throughout the afternoon and evening. The data was then used to analyze park engagement relative to time and location and visualized on a time-series basis.

Adafruit LED Strip (WSw2801)



The LED strip has a 25 pixels series. Each pixel uses an 8mm diffused RGB LEDs, with a 120-degree beam width. The total max brightness of all LEDs is about 1600mcd. Each pixel is digitally controlled, with an internal 8-bit PWM LED driver (24-bit color for 16 million different shades). Data is shifted down from one pixel to the next and controlled in a microcontroller.

Human Presence Sensor (AK9753)

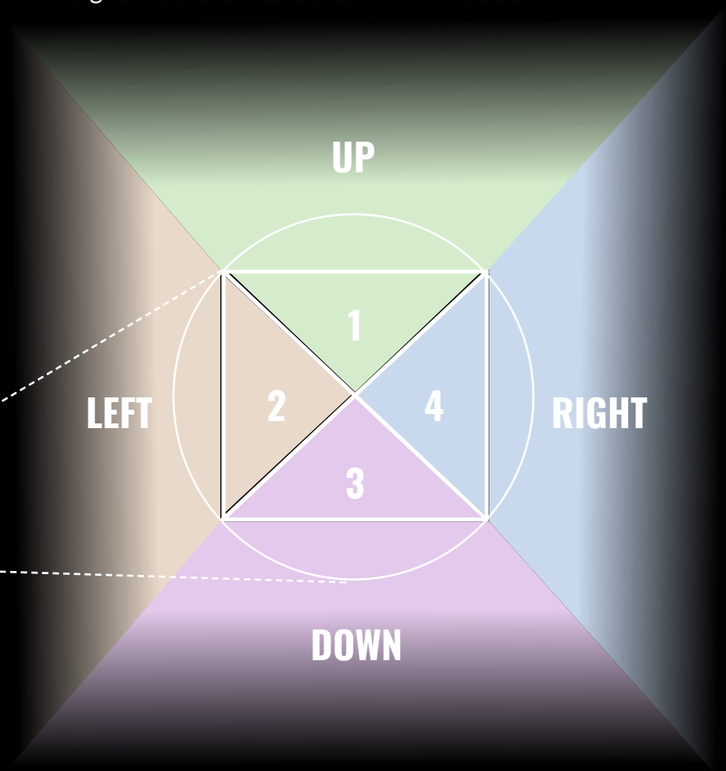


Front of the Human Presence Sensor



Back of the Human Presence Sensor

The Human Presence Sensor contains four infrared sensors facing four different directions illustrated below.

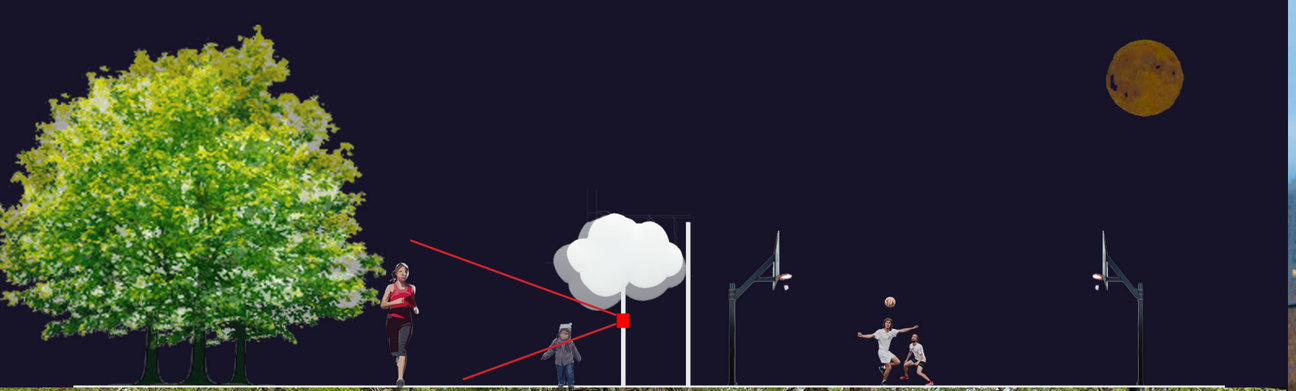


Implementation

Starlight Park, Bronx, New York
Time: May 4th, 2019, 5:15 - 8:15 pm

Starlight Park is located alongside the Sheridan Expressway, which was recently reclaimed, reconstructed, and expanded. The park has large open spaces, paved pedestrian paths, various sports facilities, and a children's playground. The prototype was deployed here on May 4th, 2019 for three hours from 5:15 to 8:15PM.



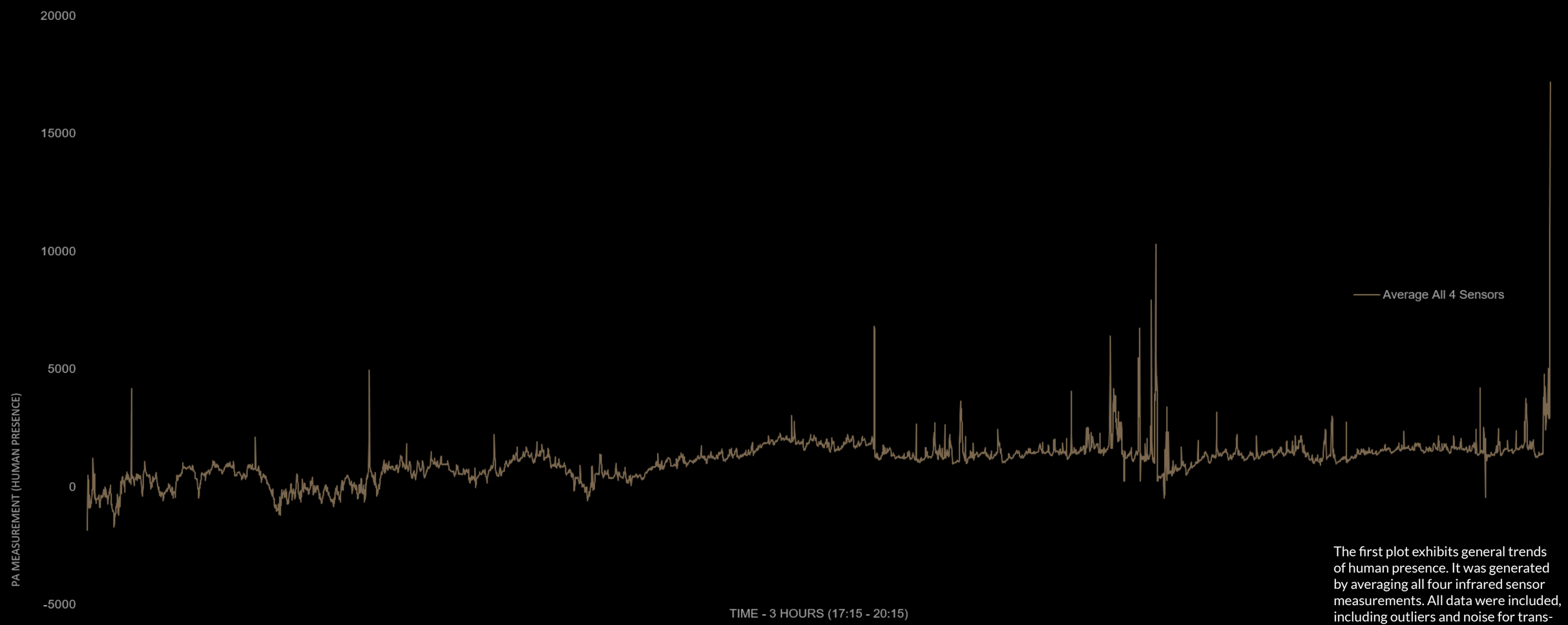






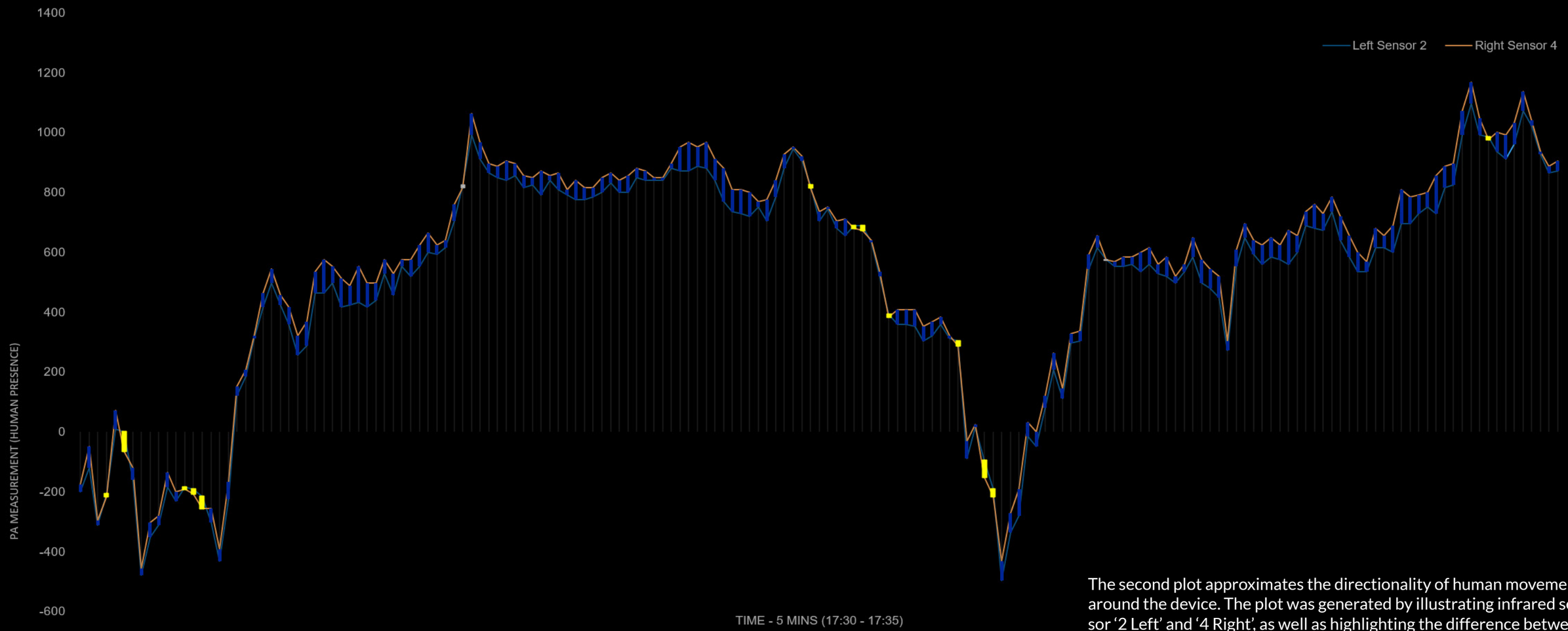
Analysis

The data collected was analyzed on a time-series basis focusing on three main areas of interests. The data analysis was focused on: 1) general human presence, 2) direction of movement, and 3) the presence of children, as a proxy for public safety.



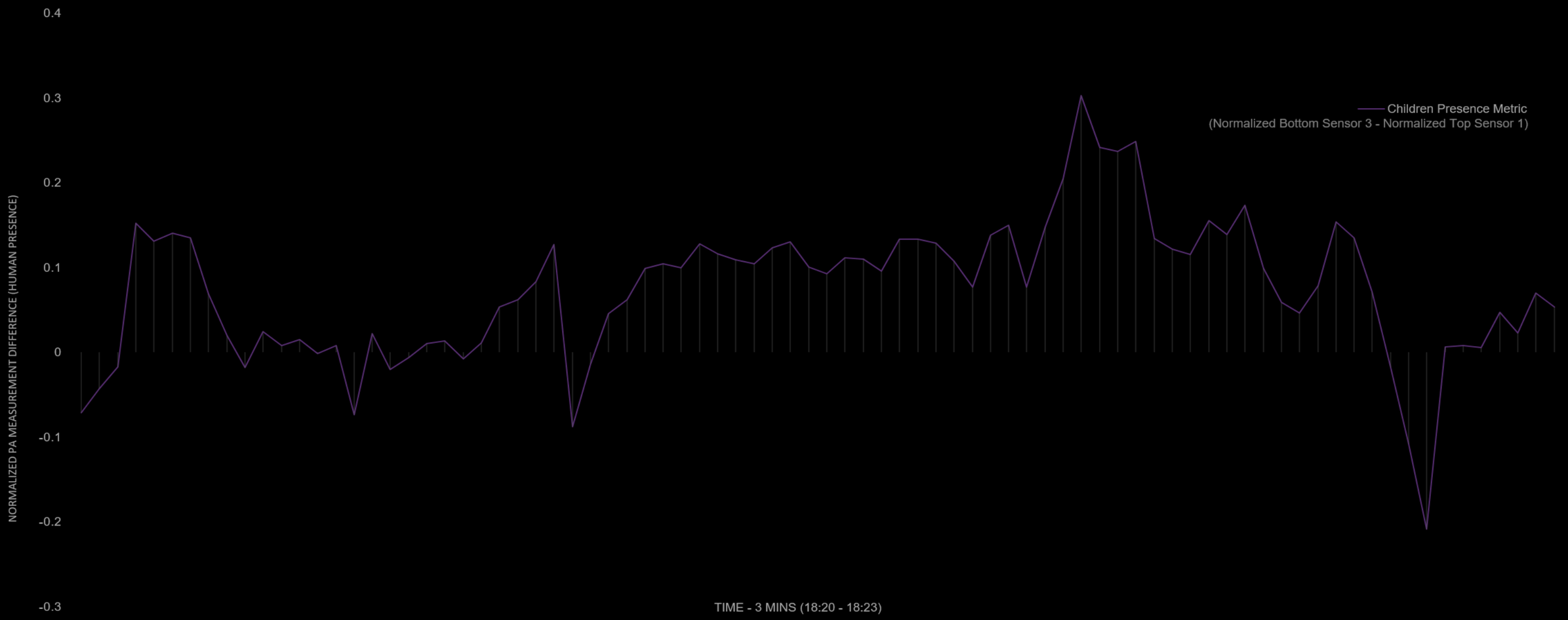
The first plot exhibits general trends of human presence. It was generated by averaging all four infrared sensor measurements. All data were included, including outliers and noise for transparency.

Analysis



The second plot approximates the directionality of human movement around the device. The plot was generated by illustrating infrared sensor '2 Left' and '4 Right', as well as highlighting the difference between them. The blue highlights indicate that there was a generally higher degree of human presence to the right of the device, while the yellow highlights indicate that there was generally a higher degree of human presence to the left of the device. Taken together, approximations of directionality could be revealed. The following scenario demonstrates human movement generally to the right of the device, which appears to have moved from right to left, perpendicular to the path it was placed on.

Analysis



The third plot approximates the presence of children by height. The plot was generated by first normalizing the values of sensor ‘1 Top’ and ‘3 Bottom’. Second, the difference between the normalized values was calculated by subtracting ‘3 Bottom’ from ‘1 Top’. The higher the difference between the normalized values of the two sensors would indicate the presence of a child. The presence of children was then used to infer what that might mean, as a metric for public safety.

Scalability

Cloudy Night was envisioned to be able to scale in any urban environment where lighting is necessary or augmented illuminance would enhance urban experiences. It was thought that urban environments that could benefit interactive feedback loops might take form along linear pathways frequented by pedestrian foot traffic such as urban sidewalks, public park trails, walking paths, or pedestrian malls. The prototype has demonstrated the viability of the idea. However, future implementation might likely benefit from multiple devices. Furthermore, future scalability might be improved by analyzing the data comprehensively to create a more complete understanding of both urban mobility and public safety.





Conclusion

Cloudy Night was a data collecting prototype to assess if interactive lighting could enhance public park engagement and safety. Through the development of the device, it was found that both directionality of human movement and the presence of children could be roughly approximated. The presence of children was used as a proxy for the perception of public safety. Although there were obvious limitations of the project such as: the technical limits of the sensors and microcontrollers (range and accuracy), the use of a single prototype, the lack of a control group in the analysis, the duration and single location of data collection, and many others, it was concluded that the Cloudy Night prototype was a useful device for illuminating low lit areas with fun, interactive, and playful feedback loops, while assessing these mediums of interaction through data collection to enhance park engagement and public safety. The project has much room for improvement. However, it was one attempt in a new and emerging field of urbanism to help create a better city for everyone.

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